

Probability Theory

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These are my notes on probability theory, which I am studying for C271A and as an extension of what I learned in analysis. I hope to cover the relevant measure theory whenever possible, and I will also add completed practice problems as well.

Textbooks Used to Develop these Notes:

1. *Probability: Theory and Examples* - Rick Durrett
2. *Measure, Integration, and Real Analysis* - Sheldon Axler

Other Textbooks Referenced:

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NOTE: Any sections/subsections marked with an asterisk are extra sections and have been added to this document as extensions of introduced theory.

1 Probability Spaces

We begin with some basic measure theory which will motivate the formulation of the σ -algebra and allow us to go from generality to specificity in defining probability spaces.

We begin by defining a probability space as a triple $(\Omega, \mathcal{F}, P)$. The first element Ω is called the set of outcomes or the "sample space", which is just an underlying set that can be defined as we wish.

Definition 1.1 (σ -algebra). *The second element $\mathcal{F}$ is called a σ -algebra, satisfying the following properties:*

1. $\Omega \in \mathcal{F}$
2. $\forall S \in \mathcal{F} : S \subseteq \Omega \wedge \Omega \setminus S \in \mathcal{F}$
3. *The countable (finite or countably infinite) union of sets in $\mathcal{F}$ is in $\mathcal{F}$:*

$$\forall \{S_n\}_{n \in \mathbb{N}} \in \mathcal{F}^{\mathbb{N}} : \bigcup_{n \in \mathbb{N}} S_n \in \mathcal{F}$$

We call the set S an event (specifically an event with an assigned probability, which we shall see later).

Lemma 1.1. *We can derive other properties of sets in the σ -algebra, such as:*

1. $\emptyset \in \mathcal{F}$
2. $\forall \{S_n\}_{n \in \mathbb{N}} \in \mathcal{F}^{\mathbb{N}}$:

$$\bigcap_{n \in \mathbb{N}} S_n \in \mathcal{F}$$

3. $\forall S, T \in \mathcal{F} : S \setminus T \in \mathcal{F}$

Proof. For $\emptyset \in \mathcal{F}$, this follows from the fact that $\Omega \in \mathcal{F}$ and the complement of Ω then must be in $\mathcal{F}$.

The second statement follows from DeMorgan's Laws. If $S_n \in \mathcal{F}$, then $S_n^c \in \mathcal{F}$, which tells us that:

$$\bigcup_{n \in \mathbb{N}} S_n^c = \left(\bigcap_{n \in \mathbb{N}} S_n \right)^c \in \mathcal{F}$$

Once again by the complement axiom, we see that:

$$\bigcap_{n \in \mathbb{N}} S_n \in \mathcal{F}$$

As for the third statement, we know that $S \setminus T = S \cap (\Omega \setminus T)$. Since $T \in \mathcal{F}$, then $\Omega \setminus T \in \mathcal{F}$, so by statement 2, we have that statement 3 follows. $\square$

Lemma 1.2. *The intersection of a family of σ -algebras for a set Ω is a σ -algebra.*

Proof. Let us call the family $\{\mathcal{F}_\alpha : \alpha \in I\}$ such that we have valid measurable spaces $(\Omega, \mathcal{F}_\alpha)$, then:

$$\mathcal{F} := \bigcap_{\alpha \in I} \mathcal{F}_\alpha$$

is a valid sigma algebra such that $(\Omega, \mathcal{F})$ is a measurable space. We see that:

$$\forall \alpha \in I : \Omega \in \mathcal{F}_\alpha \implies \Omega \in \bigcap_{\alpha \in I} \mathcal{F}_\alpha$$

so property 1 in Definition 1.1 is fulfilled. Now, for some S in the intersection:

$$\begin{aligned} S \in \bigcap_{\alpha \in I} \mathcal{F}_\alpha &\iff \forall \alpha \in I : S \in \mathcal{F}_\alpha \\ &\implies \forall \alpha \in I : S^c \in \mathcal{F}_\alpha \iff S^c \in \bigcap_{\alpha \in I} \mathcal{F}_\alpha \end{aligned}$$

thus property 2 is fulfilled. Now consider some $\{S_n\}_{n \in \mathbb{N}} \in \mathcal{F}^{\mathbb{N}}$, then:

$$\begin{aligned} \forall n \in \mathbb{N} : S_n \in \mathcal{F} &\implies \forall \alpha \in I : S_n \in \mathcal{F}_\alpha \\ \implies \forall \alpha \in I : \{S_n\}_{n \in \mathbb{N}} \in \mathcal{F}_\alpha^{\mathbb{N}} &\implies \forall \alpha \in I : \bigcup_{n \in \mathbb{N}} S_n \in \mathcal{F}_\alpha \\ &\implies \bigcup_{n \in \mathbb{N}} S_n \in \mathcal{F} \end{aligned}$$

so we have verified property 3. This is quite similar to topologies, where the intersection of topologies is a topology, but the union is not necessarily a topology. $\square$

Consider some sample space $\Omega := \{1, 2, 3, 4\}$, then consider:

$$\mathcal{F}_1 := \{\Omega, \emptyset, \{1\}, \{2, 3, 4\}\}$$

$$\mathcal{F}_2 := \{\Omega, \emptyset, \{2\}, \{1, 3, 4\}\}$$

both of which are σ -algebras. If we take the intersection:

$$\mathcal{F}_3 := \mathcal{F}_1 \cap \mathcal{F}_2 = \{\Omega, \emptyset\}$$

which is the "coarsest" or trivial σ -algebra. Now taking the union:

$$\mathcal{F}_1 \cup \mathcal{F}_2 = \{\Omega, \emptyset, \{1\}, \{2\}, \{2, 3, 4\}, \{1, 3, 4\}\}$$

but by Definition 1.1 property 3, we would need $\{1, 2\}$ to belong in the union, thus the union is not necessarily a σ -algebra.

1.1 Motivating the σ

The third and final element of the triple is P , which we call the probability measure. A measure is defined as a following:

Definition 1.2. $(X, \mathcal{F})$ is a measurable space on which we can put a measure $\mu : \mathcal{F} \rightarrow [0, \infty]$ with the properties:

1. $\forall S \in \mathcal{F} : \mu(S) \geq 0 \wedge \mu(\emptyset) = 0$

Note that we could also replace the second part with $\mu(\emptyset) \in [0, \infty)$.

2. Given some $\{S_n\}_{n \in \mathbb{N}} \in \mathcal{F}^{\mathbb{N}}$, if:

$$\forall i \neq j : S_i \cap S_j = \emptyset \implies \mu\left(\bigcup_{n \in \mathbb{N}} S_n\right) = \sum_{n \in \mathbb{N}} \mu(S_n)$$

This second property will be quite important, and we shall call it countable additivity. Countable additivity gives a reason as to why we wanted property 3 in Definition 1.1, where:

$$\forall \{S_n\}_{n \in \mathbb{N}} \in \mathcal{F}^{\mathbb{N}} : \bigcup_{n \in \mathbb{N}} S_n \in \mathcal{F}$$

If this were not true, then we could have had some sequence of S_n in the σ -algebra where their countably infinite union is not measurable, so property 2 in this definition would not always hold.

Lemma 1.3. This definition of a measure reveals additional properties:

1. Monotonicity:

$$\forall A, B \in \mathcal{F} : A \subseteq B \implies \mu(A) \leq \mu(B)$$

2. Subadditivity:

$$\forall \{S_n\}_{n \in \mathbb{N}} \in \mathcal{F}^{\mathbb{N}} : \mu\left(\bigcup_{n \in \mathbb{N}} S_n\right) \leq \sum_{n \in \mathbb{N}} \mu(S_n)$$

3. Continuity from below:

$$\forall n \in \mathbb{N} : S_n \subseteq S_{n+1} \implies \mu\left(\bigcup_{n \in \mathbb{N}} S_n\right) = \lim_{n \rightarrow \infty} \mu(S_n)$$

4. Continuity from above:

$$\forall n \in \mathbb{N} : S_{n+1} \subseteq S_n \wedge \mu(S_0) < \infty \implies \mu\left(\bigcap_{n \in \mathbb{N}} S_n\right) = \lim_{n \rightarrow \infty} \mu(S_n)$$

Proof. Take the sets $A, B \setminus A$, then:

$$\mu(B) = \mu(A \cup (B \setminus A)) = \mu(A) + \mu(B \setminus A) \geq \mu(A)$$

thus proving the first property of non-negativity. Here we used countable additivity with the sequence $A, B \setminus A, \emptyset, \emptyset, \dots$ to bring about finite additivity. This means the proof also requires that the measure $\mu(\emptyset) = 0$.

For the second property, we can rewrite the union as the following:

$$\bigcup_{n \in \mathbb{N}} S_n = \bigcup_{n \in \mathbb{N}} \left(S_n \setminus \bigcup_{i=0}^{n-1} S_i \right)$$

The reverse inclusion is trivial. For the forward inclusion, if x is in the union of S_n , then:

$$\exists n \in \mathbb{N} : x \in S_n$$

but it might be that x belongs to the intersection of multiple such sets, so by well ordering property and nonemptiness above, we can define:

$$m := \min\{n \in \mathbb{N} : x \in S_n\}$$

So $x \in S_m$ which also tells us that for all $n \leq m$ that $x \in S_n$, so:

$$\exists m \in \mathbb{N} : x \in S_m \setminus \bigcup_{i=0}^{m-1} S_i$$

We also have the added benefit that each set difference is disjoint from any other. So:

$$\mu\left(\bigcup_{n \in \mathbb{N}} S_n\right) = \mu\left(\bigcup_{n \in \mathbb{N}} \left(S_n \setminus \bigcup_{i=0}^{n-1} S_i\right)\right) = \sum_{n \in \mathbb{N}} \mu\left(S_n \setminus \bigcup_{i=0}^{n-1} S_i\right) \leq \sum_{n \in \mathbb{N}} \mu(S_n)$$

Now observe we can do the following rewrite:

$$\sum_{n \in \mathbb{N}} \mu\left(S_n \setminus \bigcup_{i=0}^{n-1} S_i\right) = \lim_{n \rightarrow \infty} \sum_{i=0}^n \mu\left(S_i \setminus \bigcup_{j=0}^{i-1} S_j\right)$$

Countable additivity gives us finite additivity (simply consider a sequence $A_0, A_1, A_2, \dots, A_n, \emptyset, \emptyset, \dots$), so we can rewrite the sum as:

$$= \lim_{n \rightarrow \infty} \mu\left(\bigcup_{i=0}^n S_i\right) = \mu\left(\lim_{n \rightarrow \infty} \bigcup_{i=0}^n S_i\right)$$

The reason why the third and fourth properties are dubbed "continuity" is because they essentially explain a limit exchange under certain sufficient conditions, where:

$$\mu\left(\lim_{n \rightarrow \infty} \bigcup_{i=0}^n S_i\right) = \mu\left(\lim_{n \rightarrow \infty} S_n\right) = \lim_{n \rightarrow \infty} \mu(S_n)$$

The proof follows simply from our observation before. Due to the nesting where $S_0 \subseteq S_1 \subseteq \dots$, we have that:

$$\bigcup_{i=0}^n S_i = S_n$$

By the monotonicity property, we have that $\{\mu(S_n)\}_{n \in \mathbb{N}}$ is a non-decreasing sequence, so we can rewrite the limit as:

$$\lim_{n \rightarrow \infty} \mu(S_n) = \sup_{n \in \mathbb{N}} \mu(S_n)$$

For the fourth property, we use continuity from below. Consider a sequence $S_0 \supseteq S_1 \supseteq \dots$. Then we have that $S_0 \setminus S_1 \subseteq S_0 \setminus S_2 \subseteq \dots$, which tells us that:

$$\mu\left(\lim_{n \rightarrow \infty} S_0 \setminus S_n\right) = \lim_{n \rightarrow \infty} \mu(S_0 \setminus S_n)$$

Now observe:

$$\begin{aligned} \mu(S_0) &= \mu(S_0 \setminus S_n \cup S_n) = \mu(S_0 \setminus S_n) + \mu(S_n) \implies \mu(S_0) - \lim_{n \rightarrow \infty} \mu(S_n) = \lim_{n \rightarrow \infty} \mu(S_0 \setminus S_n) \\ \implies \mu(S_0) - \lim_{n \rightarrow \infty} \mu(S_n) &= \mu(S_0 \setminus \bigcap_{n \in \mathbb{N}} S_n) = \mu(S_0) - \mu\left(\bigcap_{n \in \mathbb{N}} S_n\right) \end{aligned}$$

and via cancellation we have the desired property. Notice that the argument requires $\mu(S_0) < \infty$. If it were not less than infinity, then it may be possible that even the subsets of S_0 have infinite measure, which would mean:

$$\lim_{n \rightarrow \infty} \mu(S_n) = \infty$$

which would give an indeterminate subtraction on the left hand side. □

Furthermore, we can provide an equivalent definition for measure which does not require countable additivity:

Lemma 1.4. *The function μ is a measure if and only if it is non-negative, finitely additive, and continuous from below.*

Proof. $\implies$: This direction was proved in Lemma 1.3.

$\impliedby$: Consider a sequence of pairwise disjoint sets $A_1, A_2, \dots$. Then we have the sequence:

$$\left\{ \bigcup_{i=0}^n A_i \right\}_{n \in \mathbb{N}}$$

which is non-decreasing (satisfying our condition for continuity from below). Then:

$$\mu \left(\bigcup_{n \in \mathbb{N}} A_n \right) = \mu \left(\bigcup_{n \in \mathbb{N}} \left(\bigcup_{i=0}^n A_i \right) \right) = \lim_{n \rightarrow \infty} \mu \left(\bigcup_{i=0}^n A_i \right)$$

By finite additivity, because the sequence is pairwise disjoint, we have that:

$$= \lim_{n \rightarrow \infty} \sum_{i=0}^n \mu(A_i) = \sum_{n \in \mathbb{N}} \mu(A_n)$$

Now we need to prove $\mu(\emptyset) = 0$. By finite additivity:

$$\mu(\emptyset \cup \emptyset) = \mu(\emptyset) = \mu(\emptyset) + \mu(\emptyset) \implies \mu(\emptyset) = 0$$

So we must assume non-negativity (if we try to prove it via monotonicity, it becomes circular). Then we have that:

$$\text{finitely additive} + \text{continuous from below} \iff (\mu(\emptyset) = 0) + \text{countably additive}$$

□

We can equivalently define a measure as:

Definition 1.3 (Alternative Definition of Measure). *A measure $\mu : \mathcal{F} \rightarrow [0, \infty]$ has the properties:*

1. *For all finite collections of pairwise disjoint sets $S_0, S_1, \dots, S_n$:*

$$\mu \left(\bigcup_{i=0}^n S_i \right) = \sum_{i=0}^n \mu(S_i)$$

2. *For all sequences $\{S_n\}_{n \in \mathbb{N}} \in \mathcal{F}^{\mathbb{N}}$:*

$$\forall n \in \mathbb{N} : S_n \subseteq S_{n+1} \implies \mu \left(\bigcup_{n \in \mathbb{N}} S_n \right) = \lim_{n \rightarrow \infty} \mu(S_n)$$

We can now make this notion more specific to probability and say:

Definition 1.4. *A probability measure $P : \mathcal{F} \rightarrow [0, 1]$ is one where $P(\Omega) = 1$.*

To motivate the use of σ -algebras, we look at a simple probability based example. Consider a weighted coin where the conductor of the experiment is not aware of the weightage. Then:

$$\Omega := \{H, T\}$$

However, our σ -algebra would be:

$$\mathcal{F} := \{\Omega, \emptyset\}$$

because $P(\{H\})$ or $P(\{T\})$ would not be known. We can make a more complicated example.

we fix our attention to subsets of $\mathbb{R}$. We defined the notion of outer measure of subsets of $\mathbb{R}$ as the following:

$$\forall A \subseteq \mathbb{R} : |A| := \inf \left\{ \sum_{n \in \mathbb{N}} |I_n| : A \subseteq \bigcup_{n \in \mathbb{N}} I_n \right\}$$

where I_n is an open interval (or closed works as well) whose measure we defined as the difference between its endpoints $b - a$.